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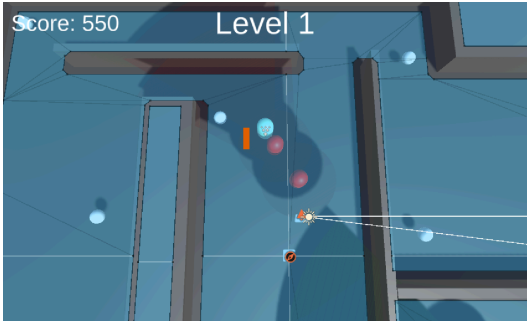
Shawn Kramer — skkramer@ucsc.edu — Github: “Shawn-Kramer”

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Stealth Game Expansion

Additions/changes:

- 2 new stages for a total of 3 levels, which get progressively more difficult
- Added sprint while holding shift
 - Particles in the opposite direction of running.
- Security cameras
- Main menu
 - Start, settings (music/sfx volume sliders), credits, quit buttons
- Custom shader materials
 - Created a scan line texture
 - Used a shader graph to create a hologram-like material
 - The material was used for the guard’s vision cones
- Guard Alert Chain Mechanic
 - Once alerted, a guard will alert the other guards in range
- Audio
 - Looping music track
 - [Stealth Action Suspense by FilyTunes - Unity Asset Store](#)
 - Sfx for collecting collectibles, reaching goal, being detected, and being caught
 - [FREE Casual Game SFX Pack by Dustyroom - Unity Asset Store](#)
- Collectibles and scoring system



Reflections:

Bryce Han:

For this project, I primarily focused on visual feedback and AI coordination systems. One of my main contributions was converting the guard vision cones from the transparent meshes to a custom hologram shader. This required me to learn how Shader Graph materials interacted with generated meshes. Another addition I made was the alert chain mechanic, this allowed nearby guards to communicate player detection.

The biggest challenge, *by far*, was integrating the hologram shader with the vision cone system. This meant that I had to refactor the vision cone logic in the guardController scripts and troubleshooting issues such as missing scanlines, incorrect transparency behavior, and color values not displaying as expected taught me a great deal about how shaders process color and alpha information. I still think it could look better. However, currently it's serviceable. Generally, these hurdles helped me understand Unity's render pipeline and how visual effects can be driven by gameplay systems.

Shawn Kramer:

My contributions were on the construction of the big level and the main menu. Throughout this, the biggest technical challenges I encountered were related to some GitHub conflicts and setting up the Unity UI packages. Ivan and Bryce helped me resolve these as they came up, and overall, this experience had the least conflict of any of the Unity projects I have worked on.

Over the quarter, I went from not knowing anything about Unity to being able to navigate the editor and do many of the important basics, such as setting up a GitHub repository, creating objects, tweaking them in the inspector, and attaching scripts to them, as well as knowing how to figure out things I don't yet know to continue improving. If I had more time, I would have liked to add more detail or gameplay depth to some of the large spaces of the map, or introduce new mechanics in future levels. Otherwise, I could fine-tune the difficulty more closely to create a better experience.

Ivan Li:

My primary contributions to this project were all the audio handling/integration (including volume sliders), implementation of security cameras, implementation of the sprint mechanic and its sprint bar UI, persistent UI for score and level through scenes, and implementation of the collectibles and scoring system. The only issues/challenges I ran into were creating some of the UI in a 3D space, which wasn't too bad once I figured it out, but was a bit annoying when I was trying to create a sprint bar visual that's attached to the player. After that

however, the other UI was not a challenge, especially since it was just on screen and not correlated to any objects. Overall, it was not the most challenging process, just a bit tedious to create new levels and adding all the new stuff.

In terms of what I learned, there isn't much to say, since I've gained quite a bit of Unity experience throughout my quarter, especially through CMPM 170 Rapid Prototyping. For example, the AudioManager system I used in this project was pulled from prior projects in which I created and iterated upon to get to this system which makes music and sound effect handling very simple. For the amount of time allotted, I think I made a pretty substantial amount of progress, but some things I might've done differently were cleaning up the UI (maybe with imported fonts and actual rich text tags to make things look better), and being more organized in general (eg: I forgot to make the Security Cameras a prefab).